



2.7 Polynomial functions and factor theorem (HL)

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The factor theorem states that $(x-k)$ is a factor of polynomial $P(x)$ if and only if $P(k)=0$. The remainder theorem states that dividing $P(x)$ by $(x-k)$ leaves remainder $P(k)$. These allow factorisation of higher-degree polynomials by testing possible roots, then polynomial division. For a polynomial with roots $\alpha, \beta, \gamma, \dots$, Vieta's formulas relate the sum and products of roots directly to the coefficients.

Worked example: follow the reasoning

Factorise $P(x) = x^3 - 6x^2 + 11x - 6$ completely.
Test $x=1$: $P(1)=1-6+11-6=0$, so $(x-1)$ is a factor.
Divide: $x^3-6x^2+11x-6 = (x-1)(x^2-5x+6) = (x-1)(x-2)(x-3)$.

Quick check · multiple choice

If $P(x)=x^3+2x^2-5x-6$ and $(x+1)$ is a factor, what is $P(-1)$?

- A -6
- B 0
- C 2
- D 6

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Consider $P(x) = 2x^3 - 3x^2 - 11x + 6$. [7]
- (a) Show that $(x-3)$ is a factor of $P(x)$.
 - (b) Hence factorise $P(x)$ completely and find all real roots.
 - (c) For the cubic $x^3+px+q=0$ with roots α, β, γ , state (without proof) the value of $\alpha+\beta+\gamma$ in terms of p and q .

Final answer: _____



Answers and explanation

Question 1 [7]

- (a) $P(3) = 2(27) - 3(9) - 11(3) + 6 = 54 - 27 - 33 + 6 = 0$, so $(x-3)$ is a factor.
- (b) Dividing: $2x^3 - 3x^2 - 11x + 6 = (x-3)(2x^2 + 3x - 2) = (x-3)(2x-1)(x+2)$. Roots: $x=3$, $x=1/2$, $x=-2$.
- (c) For $x^3 + px + q = 0$ (no x^2 term, so coefficient of x^2 is 0), $\alpha + \beta + \gamma = -0/1 = 0$.

Quick check answer

B. By the factor theorem, if $(x+1)$ is a factor then $P(-1)=0$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.